

TrES-1b

R-0182

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_180420 · created 2026-07-16T19:23:27+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458228.8403 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1422 +/- 0.0091
Transit depth (Rp/Rs)^2	2.02 +/- 0.26 [%]
Orbital Inclination (inc)	89.86 +/- 0.65
Airmass coefficient 1 (a1)	1.003 +/- 0.012
Airmass coefficient 2 (a2)	-0.0027 +/- 0.0019
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#2 - [224, 87]
Optimal Aperture	4.35
Optimal Annulus	8.59
Transit Duration (day)	0.1043 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1422 ± 0.0091 vs 0.1358 (deviation 0.7σ)
Duration fit vs published	0.1043 d vs 0.1042 d (deviation 0.11σ)
Mid-time O–C	-4.6 min
Depth SNR	15.6
Residual scatter	1.2 %

Light curve

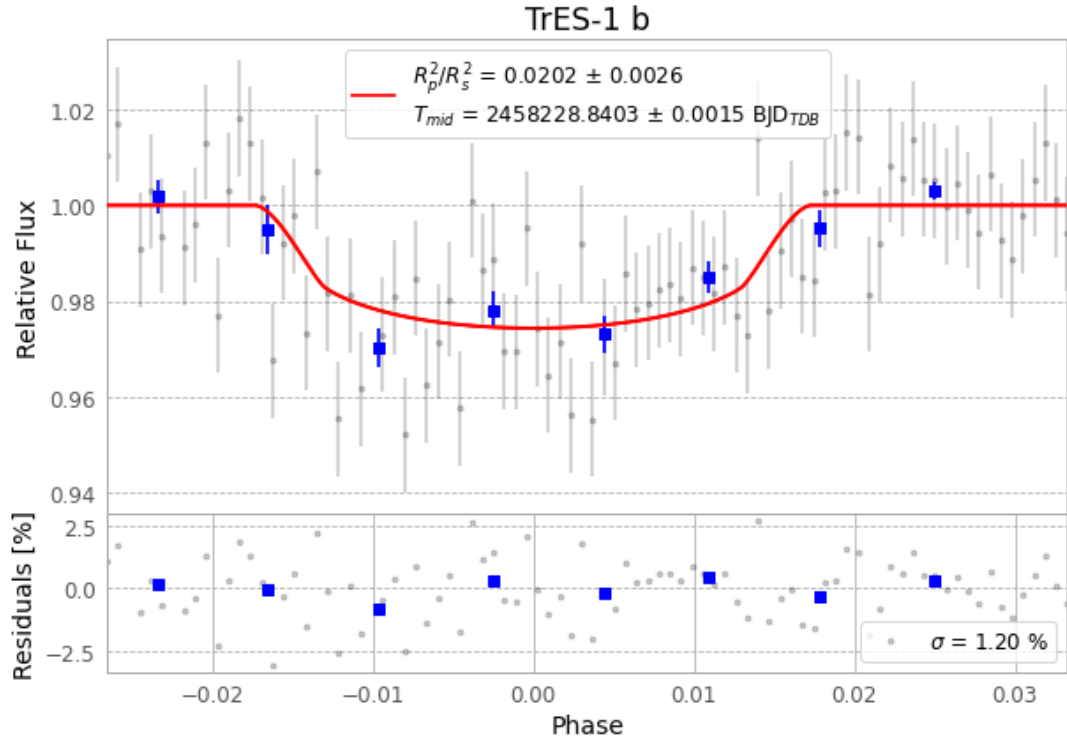


Figure 1 — FinalLightCurve_TrES-1 b_2018-04-19.png (SHA-256 ca18c35c22a0384a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 142], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 92009a40313c7b7e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca

Data provenance

88 FITS frames (58 MB), TRES-1180420061212.FITS → TRES-1180420103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-180420.log (c23c8f7786e7...), FinalLightCurve_TrES-1 b_2018-04-19.png (ca18c35c22a0...), FinalLightCurve_TrES-1 b_2018-04-19.csv (0d6978438ae2...)

Compute resources

Wall time 185.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:23Z.